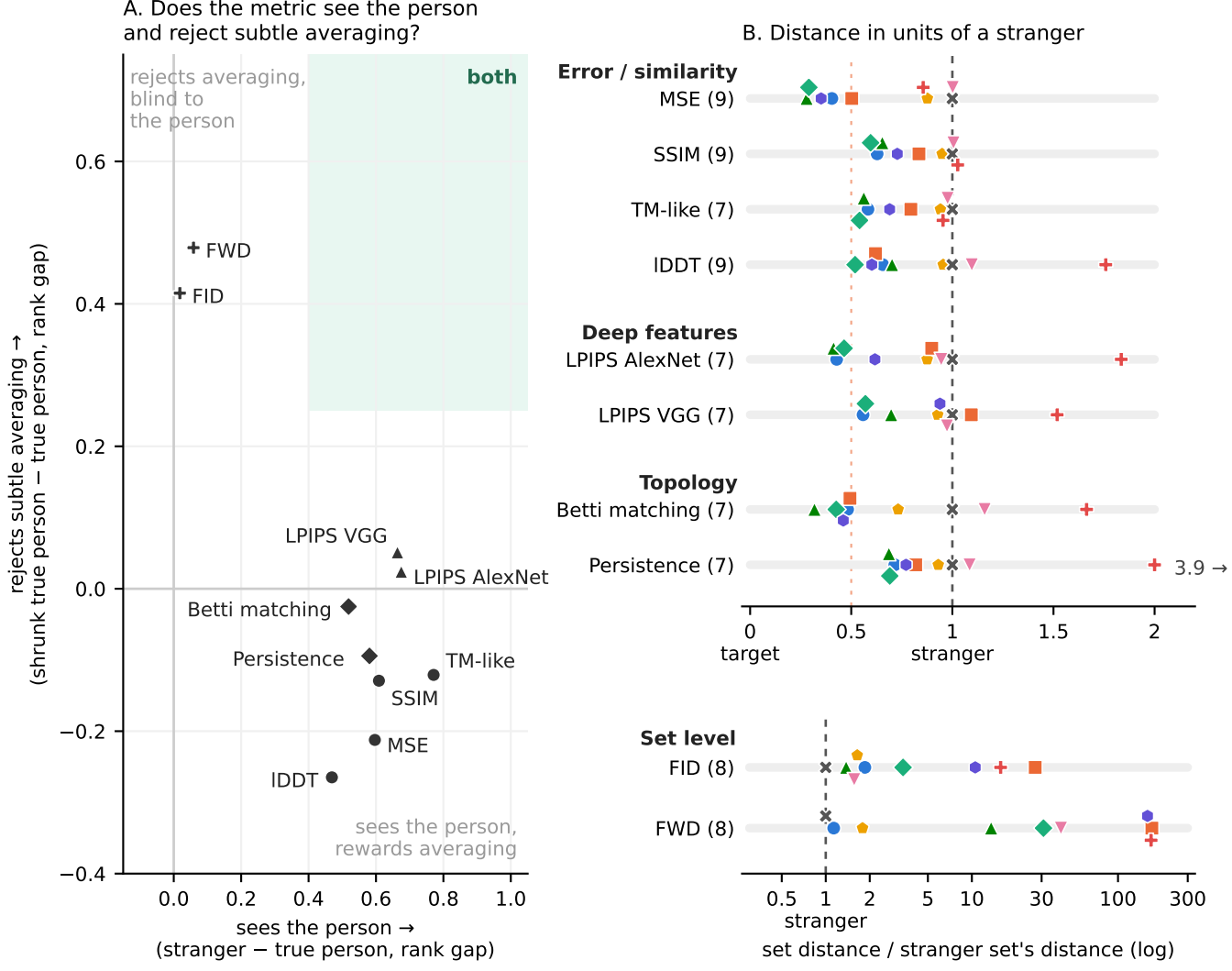
A. Does the metric see the person and reject subtle averaging?
rejects averaging, blind to the person
both
sees the person, rewards averaging
FWD
FID
LPIPS VGG
LPIPS AlexNet
Betti matching
Persistence
TM-like
SSIM
MSE
IDDT
rejects subtle averaging →
(shrunk true person − true person, rank gap)
sees the person →
(stranger − true person, rank gap)
B. Distance in units of a stranger
Error / similarity
MSE (9)
SSIM (9)
TM-like (7)
lDDT (9)
Deep features
LPIPS AlexNet (7)
LPIPS VGG (7)
Topology
Betti matching (7)
Persistence (7)
3.9 →
target
stranger
Set level
FID (8)
FWD (8)
stranger
set distance / stranger set's distance (log)
Ladder:
true person
oracle look-alike*
sibling
dataset mean
stranger
Checks:
shrunk true person
mean of k look-alikes*
wrong detail
white noise